1.1 Proof

What is a mathematical proof? A proof is a logical argument that establishes the truth of a mathematical statement beyond all doubt. Unlike a demonstration or example, a proof must work for all cases covered by the statement.

Types of Proof

Type	Method	When to Use
Deduction	Start from known facts and use logical steps to reach the conclusion	General statements about all numbers/cases
Exhaustion	Check every possible case individually	Finite number of cases
Counter-example	Find one case where the statement fails	Disproving a statement
Contradiction	Assume the opposite is true and show this leads to an impossibility	Difficult to prove directly; irrational numbers, infinite sets

Choosing the Right Proof Method

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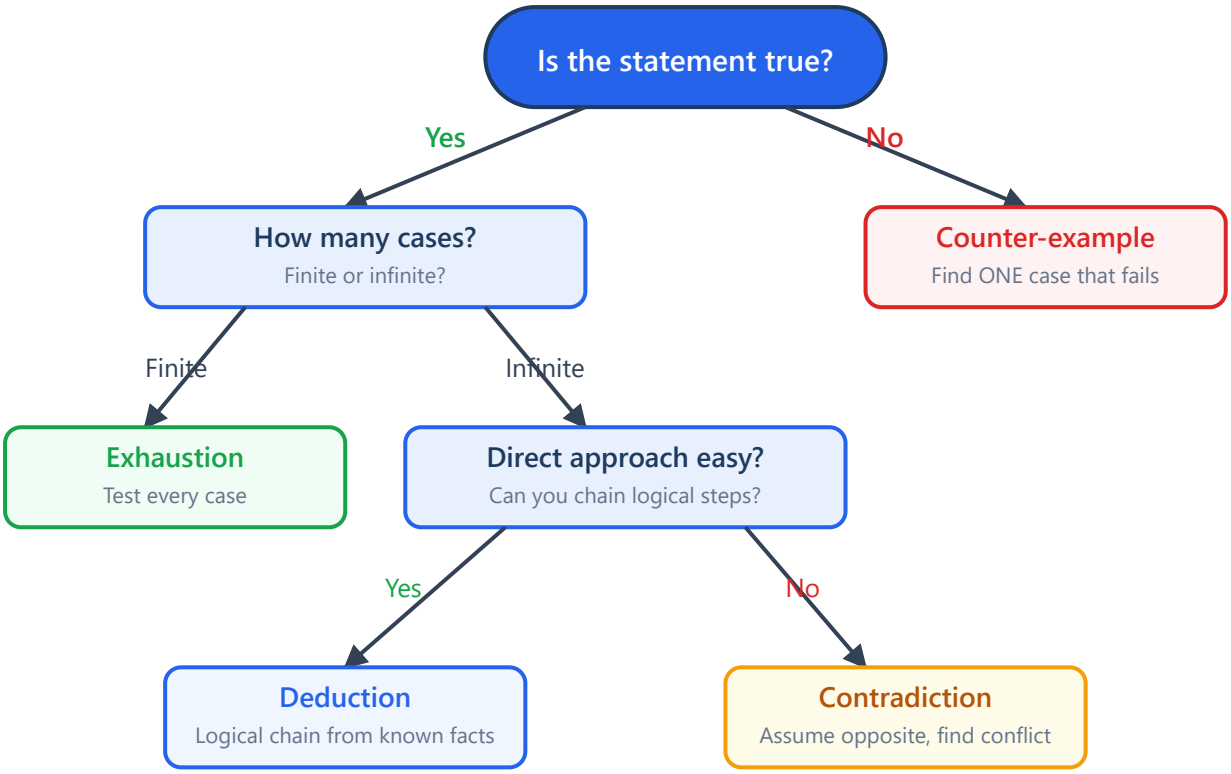


Proof by Deduction

Proof by deduction starts from known mathematical facts (axioms, definitions, previously proven theorems) and uses logical steps to arrive at the required conclusion. This is the most common type of proof.

Key Algebraic Representations

- An even number can be written as $2n$ where $n \in \mathbb{Z}$
- An odd number can be written as $2n + 1$ where $n \in \mathbb{Z}$
- Consecutive integers: $n, n + 1, n + 2, \ldots$
- A multiple of 3: $3n$ where $n \in \mathbb{Z}$
- A perfect square: n^2 where $n \in \mathbb{Z}$

Example 1: Prove that the sum of two consecutive odd numbers is divisible by 4.

Let the two consecutive odd numbers be $2n+1$ and $2n+3$ where $n \in \mathbb{Z}$.

Sum = $(2n+1)+(2n+3) = 4n+4 = 4(n+1)$

Since $n+1$ is an integer, $4(n+1)$ is divisible by 4. $\square$

Example 2: Prove that the product of two odd numbers is always odd.

Let the two odd numbers be $2a+1$ and $2b+1$ where $a, b \in \mathbb{Z}$.

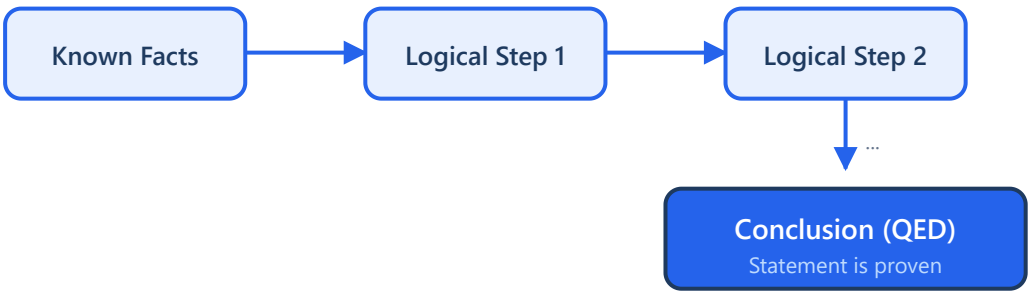
Product = $(2a+1)(2b+1) = 4ab+2a+2b+1 = 2(2ab+a+b)+1$

Since $2ab+a+b$ is an integer, this is of the form $2k+1$, which is odd. $\square$

Example 3: Prove that n^2+n is even for all $n \in \mathbb{Z}$.

Factorise: $n^2+n = n(n+1)$

This is the product of two consecutive integers. One of them must be even, so the product is even. $\square$



Proof by Exhaustion

Proof by exhaustion involves checking every possible case to confirm the statement is true. This is only practical when there are a finite (and manageable) number of cases.

Example 4: Prove that $n^2 + 2$ is not divisible by 4 for any integer n.

Any integer n is either even or odd. We test both cases:

Case 1: n is even, so $n = 2k$.
Then $n^2 + 2 = 4k^2 + 2 = 2(2k^2 + 1)$.
Since $2k^2 + 1$ is odd, this equals $2 \times \text{odd}$, which is not divisible by 4.

Case 2: n is odd, so $n = 2k + 1$.
Then $n^2 + 2 = 4k^2 + 4k + 1 + 2 = 4k^2 + 4k + 3 = 4(k^2 + k) + 3$.
This leaves remainder 3 when divided by 4, so it is not divisible by 4.

All cases checked. $\square$

Example 5: Prove that for $n \in \{1, 2, 3, 4, 5\}$, $n^2 + n + 41$ is prime.

n	$n^2 + n + 41$	Prime?
1	43	Yes
2	47	Yes
3	53	Yes
4	61	Yes
5	71	Yes

All 5 cases produce primes. $\square$

Exam Tip: Make sure you clearly label each case and check every single one. Examiners look for a systematic approach, not random testing. State your cases before you begin checking them.

Proof by Counter-Example

Proof by counter-example is used to *disprove* a statement. You only need to find **one** example where the statement fails. This proves that the statement is not always true.

Example 6: Disprove: "If n is prime, then $2^n - 1$ is also prime."

Try $n = 11$: $2^{11} - 1 = 2047 = 23 \times 89$

Since 2047 is not prime, the statement is false. $\square$

Example 7: Disprove: "The sum of two irrational numbers is always irrational."

Let $a = \sqrt{2}$ and $b = -\sqrt{2}$. Both are irrational.

Sum: $a + b = \sqrt{2} + (-\sqrt{2}) = 0$, which is rational.

Counter-example found, so the statement is false. $\square$

Example 8: Disprove: "$n^2 > n$ for all integers n."

Try $n = 0$: $0^2 = 0$, and $0 \not> 0$. The statement fails. $\square$

(Also fails for $n = 1$: $1^2 = 1 \not> 1$)

Exam Tip: When disproving, you only need ONE counter-example. State it clearly, show the calculation, and explicitly say the statement is false. Try simple values first: 0, 1, -1, 2.

Proof by Contradiction

Proof by contradiction works by assuming the opposite of what you want to prove, then showing this assumption leads to a logical impossibility (a contradiction). Since mathematics cannot contain contradictions, the original assumption must be wrong, so the statement is true.

Steps for Proof by Contradiction

1. Assume the **negation** of the statement you want to prove.
2. Use logical reasoning from this assumption.
3. Arrive at a **contradiction** (something impossible or self-contradictory).
4. Conclude that the assumption was wrong, so the original statement is true.

Example 9: Prove that $\sqrt{2}$ is irrational.

Assume for contradiction that $\sqrt{2}$ is rational.

Then $\sqrt{2} = \dfrac{a}{b}$ where $a, b \in \mathbb{Z}$, $b \neq 0$, and the fraction is in its simplest form (so a and b are not both even).

Squaring both sides: $2 = \dfrac{a^2}{b^2}$, so $a^2 = 2b^2$.

This means a^2 is even, so a must be even. Write $a = 2k$.

Then $(2k)^2 = 2b^2 \Rightarrow 4k^2 = 2b^2 \Rightarrow b^2 = 2k^2$.

This means b^2 is even, so b is even.

Contradiction: Both a and b are even, but we said the fraction was in simplest form.

Therefore $\sqrt{2}$ is irrational. $\square$

Example 10: Prove that there are infinitely many prime numbers.

Assume for contradiction that there are finitely many primes: $p_1, p_2, \ldots, p_n$.

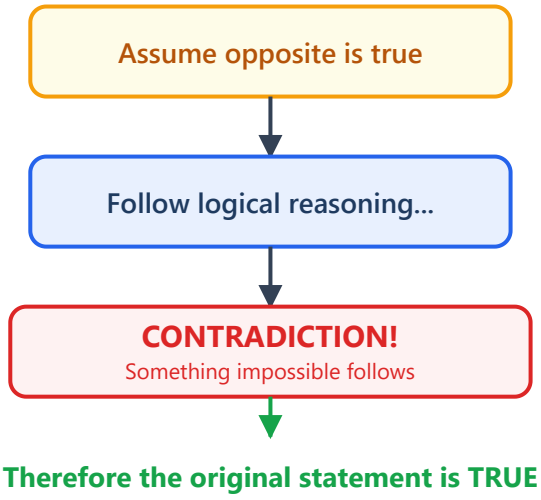
Consider the number $N = p_1 \times p_2 \times \cdots \times p_n + 1$.

N is not divisible by any of $p_1, p_2, \ldots, p_n$ (since dividing always leaves remainder 1).

So either N is itself prime (a new prime not in our list), or N has a prime factor not in our list.

Contradiction: We assumed our list contained all primes.

Therefore there are infinitely many primes. $\square$



Exam Tip: Always begin a proof by contradiction with "Assume for contradiction that..." and end by clearly stating what the contradiction is. The examiner must see you have identified the impossibility.

Common Proof Language

Symbol / Phrase	Meaning
$\therefore$	Therefore
$\because$	Because
$\square$ or QED	End of proof ("which was to be demonstrated")
$\Rightarrow$	Implies
$\Leftrightarrow$	If and only if
$\forall$	For all
$\exists$	There exists
$n \in \mathbb{Z}$	n is an integer
WLOG	Without loss of generality

Proof by Induction [Year 2 — A-Level only]

Proof by mathematical induction proves a statement is true for all positive integers n (or all integers $n \geq n_0$) by two steps:

1. **Base case:** Show the statement is true for the smallest value (usually $n = 1$).
2. **Inductive step:** Assume the statement is true for $n = k$ (the *inductive hypothesis*). Then prove it must also be true for $n = k + 1$.

Since the base case works, and each step leads to the next, the statement holds for all integers $n \geq 1$.

Induction is like toppling dominoes

Standard Result: Sum of Integers

Example 11: Prove by induction that $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$ for all $n \in \mathbb{Z}^+$.

Base case ($n = 1$): LHS $= 1$. RHS $= \frac{1 \times 2}{2} = 1$. LHS = RHS ✓

Inductive hypothesis: Assume true for $n = k$, i.e. assume $\sum_{r=1}^{k} r = \frac{k(k+1)}{2}$.

Inductive step: We need to show the result holds for $n = k + 1$, i.e. that $\sum_{r=1}^{k+1} r = \frac{(k+1)(k+2)}{2}$

.

$$\sum_{r=1}^{k+1} r = \sum_{r=1}^{k} r + (k+1) = \frac{k(k+1)}{2} + (k+1) = (k+1)\left(\frac{k}{2} + 1\right) = \frac{(k+1)(k+2)}{2}$$

This is the required form with $n = k + 1$. ✓

Conclusion: The base case holds for $n = 1$. The inductive step shows that if the result holds for $n = k$, it holds for $n = k + 1$. By the principle of mathematical induction, the result holds for all $n \in \mathbb{Z}^+$. $\square$

Induction for Divisibility

Example 12: Prove by induction that $6^n - 1$ is divisible by 5 for all $n \in \mathbb{Z}^+$.

Base case ($n = 1$): $6^1 - 1 = 5 = 5 \times 1$. Divisible by 5. ✓

Inductive hypothesis: Assume $6^k - 1$ is divisible by 5, i.e. $6^k - 1 = 5m$ for some integer m.

Inductive step: Consider $n = k + 1$: \[6^{k+1} - 1 = 6 \cdot 6^k - 1 = 6(6^k - 1) + 6 - 1 = 6(5m) + 5 = 5(6m + 1)\] Since $6m + 1 \in \mathbb{Z}$, this is divisible by 5. ✓

Conclusion: By induction, $6^n - 1$ is divisible by 5 for all $n \in \mathbb{Z}^+$. $\square$

Induction for Matrices

Example 13: Prove by induction that $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 3^n & \frac{3^n-1}{2} \\ 0 & 1 \end{pmatrix}$ for all $n \in \mathbb{Z}^+$

.

Base case ($n = 1$): LHS = $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$. RHS = $\begin{pmatrix} 3 & \frac{3-1}{2} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$. ✓

Inductive hypothesis: Assume $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}^k = \begin{pmatrix} 3^k & \frac{3^k-1}{2} \\ 0 & 1 \end{pmatrix}$.

Inductive step: \[\begin{pmatrix}3&1\\0&1\end{pmatrix}^{k+1} = \begin{pmatrix}3^k & \frac{3^k-1}{2}\\0&1\end{pmatrix}\begin{pmatrix}3&1\\0&1\end{pmatrix} = \begin{pmatrix}3^{k+1} & 3^k + \frac{3^k-1}{2}\\0&1\end{pmatrix}\] Now: $3^k + \dfrac{3^k - 1}{2} = \dfrac{2 \cdot 3^k + 3^k - 1}{2} = \dfrac{3 \cdot 3^k - 1}{2} = \dfrac{3^{k+1} - 1}{2}$. ✓

Conclusion: By induction, the result holds for all $n \in \mathbb{Z}^+$. $\square$

Proof by induction — checklist for full marks:

- State and verify the **base case** explicitly (show LHS = RHS or state "divisible by ...").
- Write out the **inductive hypothesis** clearly: "Assume true for $n = k$, i.e. ..."
- In the inductive step, start from the *k+1* expression and use the inductive hypothesis to simplify. Do not start from what you want to prove.
- End with a clear **conclusion** invoking "by the principle of mathematical induction".

- The magic phrase for divisibility proofs: write $f(k+1) = f(k) \times (\text{something}) + (\text{extra term})$ and factor out the divisor.

Exam Strategy Summary

Key exam strategies for proof questions:

- **Read the question carefully** -- "prove" means deduction/exhaustion/contradiction; "disprove" or "show by counter-example" means find a counter-example.
- **For deduction**: define variables clearly (e.g. "let $n = 2k$ where $k \in \mathbb{Z}$"), show every algebraic step, and conclude with the required form.
- **For exhaustion**: list all cases at the start, work through each systematically, tick them off.
- **For counter-example**: try small/simple numbers first. One example is sufficient.
- **For contradiction**: state your assumption clearly, work logically, and explicitly identify the contradiction.
- **For induction [Year 2]**: base case → inductive hypothesis → inductive step → conclusion. All four parts needed for full marks.
- **Always end with a concluding statement** that directly addresses the original claim.

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